Public Opinion Quarterly Advance Access published November 21, 2013

Public Opinion Quarterly

WHY DO SURVEY RESPONDENTS DISCLOSE MORE WHEN COMPUTERS ASK THE QUESTIONS?

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Abstract Self-administration of surveys has been shown to increase respondents' reporting of sensitive information, and audio computer-assisted self-interviewing (ACASI) has become the self-administration method of choice for many social surveys. The study reported here, a laboratory experiment with 235 respondents, examines why ACASI seems to promote disclosure. It compares responses in a voice-only (self-administered) interface with responses to a face-to-face (FTF) human interviewer and to two automated interviewing systems that presented animated virtual interviewers with more and less facial movement. All four modes involved the same human interviewer's voice, and the virtual interviewers' facial motion was captured from the same human interviewer who carried out the FTF interviews. For the ten questions for which FTF-ACASI mode differences (generally, more disclosure in ACASI than FTF) were observed, we compared response patterns for the virtual interviewer conditions. Disclosure for most questions was

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doi:10.1093/poq/nft038

greater under ACASI than in any of the other modes, even though the two virtual interview modes involved computerized self-administration. This suggests that the locus of FTF-ACASI effects is particularly tied to the absence of facial representation in ACASI. Additional evidence suggests that respondents' affective experience (e.g., comfort) during the interview may mediate these mode effects.

Survey respondents tend to report more socially undesirable behavior when questioned by computers than by interviewers. Population estimates of cocaine and marijuana use, for example, are greater when respondents answer anonymously via audio computer-assisted self-interviewing (ACASI)—listening to recorded questions on headphones and entering answers on a laptop, so not even the interviewer can hear—than when they answer the same questions asked by a person (e.g., Tourangeau and Smith 1996; Turner et al. 1998). Reports of the average number of opposite-sex sexual partners in the past year, in the past five years, and since the age of eighteen are similarly affected by the mode of administration: women report more partners to a computer than to an interviewer, and men report fewer (Tourangeau and Smith 1996; Tourangeau et al. 1997). In general, the evidence suggests that the increased reporting of socially undesirable behavior under computerized self-administration reflects greater response accuracy; for example, respondents' reports of their poor academic performance (low grades, dropped classes, academic probation) more closely matched official records in a web survey than in a telephone interview (Kreuter, Presser, and Tourangeau 2008).

These patterns can emerge whether questions are spoken (prerecorded voice) or textual (e.g., Tourangeau and Smith 1996; Couper, Tourangeau, and Marvin 2009). They can emerge whether the spoken questions are presented in ACASI or telephone touchtone response systems, also known as T-ACASI or interactive voice response (IVR) (e.g., Tourangeau, Couper, and Steiger 2003; Steiger and Conroy 2008); they can emerge textually in web surveys and on the laptop during what is otherwise a face-to-face interview (text-CASI). Evidence like this has led to widespread adoption of audio- and text-CASI for collecting data on almost every kind of potentially sensitive topic.

What is it about automated self-administration that leads to greater reporting of socially undesirable behaviors? Or, conversely, what is it about human interviewers that leads respondents to be less forthcoming? Any one of the many differences between human interviewers and automated interviewing systems could in principle produce FTF-ACASI disclosure differences. The purpose of the study reported here is to begin to identify which features or combinations of features are actually responsible.

Understanding the causes of FTF-ACASI mode differences is particularly important because they are not always observed. Within a single survey, some questions on a sensitive topic can demonstrate a mode difference while others do not; for example, Tourangeau and Smith (1998) observed reliably more past-year and past-month drug use with ACASI and text-CASI than with face-to-face interviews but no mode difference for lifetime drug use. And a mode difference can be found for a question in one study but not for a similar question in a different study. At present there is no systematic way to predict when a mode difference is likely to emerge—for which questions, for which populations, or for which implementations of a mode. Nor is there a systematic way to decide when to deploy computerized self-administration for those respondents, questions, or surveys for which it will be most effective. Understanding the causes of FTF-ACASI disclosure differences is an important step toward anticipating when they are likely to occur.

Potential Origins of FTF-ACASI Disclosure Differences

The general explanation for FTF-ACASI disclosure differences is that the presence of an interviewer motivates respondents to present themselves in a positive light and to avoid reporting potentially harmful information. Privately reporting socially undesirable behavior to a computer seems to be less threatening than reporting that same information to a human interviewer, and it may reduce worries about subsequent disclosure of compromising information to third parties such as family members, law enforcement, or tax offices (Aquilino 1994; Schaeffer 2000; Tourangeau and Yan 2007).

Note that this line of thinking is at odds with the pervasive notion that people are more likely to disclose embarrassing information to a nonjudgmental person with whom they have rapport; this assumption is at the heart of psychotherapeutic practice, forensic interviewing techniques, and Alfred Kinsey's in-depth interviewing methods. From this perspective, one would imagine that a human interviewer should elicit *more* reports of sensitive behaviors than a computer (to the extent that survey interviewers are perceived in the same way as Kinsey sex researchers, police interviewers, or psychotherapists). Although this is not the predominant finding in comparisons of computer and human administration, a few comparisons of different human interviewing modes (FTF versus telephone) suggest that greater interviewer presence *can* sometimes lead to more disclosure. For example, in studies of smoking (Gundersen, Delnevo, and Momperousse 2007), weight (Béland and St-Pierre 2008), and driving after drinking alcohol (Béland and St-Pierre 2008), respondents were more likely to report socially undesirable behavior or facts to an interviewer face-to-face than on the telephone, where the presence of the interviewer is less evident. It is unclear whether the greater reporting of this sensitive information to face-to-face interviewers results from greater rapport with the respondent, the respondent's awareness that the interviewer might be able to verify the answer through physical observation (smelling cigarettes, seeing body mass, etc.), both, or something else. But it does suggest that interviewer administration might sometimes lead to more disclosure than computerized self-administration: a conventional computer surely cannot detect the smell of cigarettes or the respondent's BMI.[1]

A fruitful path to more fully understanding why ACASI and other modes of self-administration promote disclosure is to decompose the interviewing situation into potentially relevant elements and systematically manipulate them. This follows the long tradition in human-computer interaction of decomposing "humanness" into features that are added incrementally to interfaces to make them more human (e.g., Sproull et al. 1996; Yee, Bailenson, and Rickertsen 2007).[2] It also further tests Tourangeau, Rips, and Rasinski's (2000, 306–7) proposal that "efforts to humanize the interfaces of computer-assisted methods of self-administration may, by creating a virtual human presence, increase the level of misreporting about sensitive topics."

Here we investigate the possibility that the differences between interviewers and ACASI systems could contribute independently, or in combination, to respondents' willingness to disclose sensitive information. As table 1 outlines, we see a number of important differences between interviewers and ACASI systems, and there are no doubt others. Any one or some combination of these differences could be responsible for mode effects in the reporting of sensitive behaviors to ACASI systems and humans. For example, the primary reason for the difference may be that ACASI systems do not have *evaluative capability*, and that respondents are averse to disclosing when there is the potential for judgment or disapproval—independent of whether the interviewer is *embodied* (FTF or virtual), has *dialogue capability*, or has *agency*.[3] Or perhaps the primary reason is that ACASI systems are not visually or verbally responsive to the respondent's answers beyond advancing to the next question when an answer has been entered, and it is the lack of interviewer responsivity that promotes willingness to disclose.[4] Or perhaps embodiment is the key: perhaps

1. If sensory technologies (e.g., Gutierrez-Osuna 2004; Person, D'Mello, and Olney 2008; Nakamoto 2013) are deployed on a larger scale, respondents' expectations of computers' capabilities could change, potentially reducing the likelihood or direction of ACASI-FTF response differences in these domains.
2. For additional background, see Preece, Rogers, and Sharp (2011) on the controversies about anthropomorphism in interaction design, and Cassell (2000) on the complexities of embodying human characteristics (e.g., improved dialogue capability, speech output and recognition, embodied human facial attributes) in interfaces.
3. See, for example, Ellingsen and Johannesson (2008) for evidence from behavioral economics that the potential for verbal evaluative feedback can affect altruistic behavior.
4. In everyday conversation, speakers are well known to be affected by how their conversational partners attend to their utterances and provide interactive feedback (for a review, see Schober and Brennan [2003]).

Table 1. Differences between Human Interviewers and Typical ACASI Systems

Dimension	Human interviewer	ACASI
Dialogue capability	Interviewer produces scripted utterances in real time in ways that will never be exactly the same for all respondents.	ACASI speech is prerecorded and each question is delivered in precisely the same way every time it is administered.
	Interviewers can, potentially, produce (scripted or unscripted) neutral probes or even clarification as needed.	ACASI systems do not generally produce probes or clarification (although in principle they could).
Embodiment	Interviewer's body and face can move and express her cognitive and affective states while speaking and while listening to answers and entering them into the computer.	ACASI systems have no bodily or facial display (see Cassell and Miller [2008], 165–66; Foucault [2009]; von der Pütten et al. [2011] for discussion of potential displays by virtual interviewing systems of the future).
	By their very presence, interviewers display both visual and auditory cues of their demographic characteristics—their age, ethnicity, gender, etc.	ACASI systems only have auditory cues of the recorded interviewer's demographics.
Perceptual capability	Interviewer can observe the *respondent's* physical characteristics and displays, and can make inferences from them (valid or not) about the respondent's cognitive and affective states, the physical condition and circumstances, and demographic characteristics that might be relevant to interpreting responses.	ACASI system cannot observe or make inferences about the respondent (though in principle more advanced systems could; see Picard [1998]).
Evaluative capability	Interviewer has the ability to judge (positively or negatively) the sensitive behaviors that respondents report in the interview.	The ACASI system cannot judge.
	Interviewer can display or hide evidence of that judgment.	ACASI system cannot display or conceal judgment.
Agency	Interviewer can act of her own accord, moving when she chooses to, and continuing to be able to act no matter what the respondent does.	ACASI system only administers the survey questions and collects the answers; it cannot form intentions to do anything else nor exercise discretion about what to do; it cannot act on its own, decide not to administer the questionnaire, etc.

respondents' willingness to disclose sensitive information is reduced by the mere presence of the interviewer's face.[5] There is certainly a long-standing practice of promoting disclosure by concealing visible evidence of one's interlocutor; e.g., in the confessional booth or on the psychoanalytic couch, although obviously those situations differ from survey interviews.

The study reported here was designed to start distinguishing between these possibilities. In the study, we compared responses to questions asked by a human interviewer to responses to questions asked by three comparable computer interfaces that were derived from attributes of the same interviewer. All interfaces used speech by the same interviewer, with recorded speech in the computer-based conditions. The ACASI interface presented questions using only recorded speech. In the two other computer conditions, an animated face presented spoken questions; facial motion was captured from recordings of the FTF human interviewer and projected onto the animated face, thus creating a "virtual interviewer." In the High Animation (HA) virtual interviewer interface, multiple channels of motion were projected onto the animated face; in the Low Animation (LA) version, only the lips moved while the rest of the face was motionless. The four interfaces differed on more than just a single feature (see below); the logic was that these comparisons would allow us to rule out or in major hypotheses about what causes ACASI-FTF disclosure differences, so that future investigation might make finer, feature-by-feature distinctions.

If disclosure is greater to all the computer interfaces than to the human interviewer, this would suggest that the primary locus of ACASI mode effects is in respondents' belief that computers cannot understand (they lack dialog and perceptual capability) or negatively evaluate (they lack evaluative capability) or intentionally disclose information,[6] and their belief that humans are qualitatively different from computers. We will call this the *Human Interviewer* hypothesis. If disclosure to any interface with a face (human or virtual interviewers) differs from the audio-only interface, this would suggest that embodiment—here, the mere presence of a face —reduces disclosure, regardless of whether it is biological or digital. We will call this the *Facial Representation* hypothesis. If disclosure is affected by the degree of facial animation—say, if disclosure were equally curtailed by a high-animation interface and a human interviewer—this would suggest that a primary locus of ACASI-FTF mode

5. Faces seem to play a crucial role in the activation of interpersonal schemas, with special status in neural representation (e.g., Kanwisher, McDermott, and Chun 1997), and in newborns' preferences (e.g., Johnson et al. 1991) as well as their cognitive capacities (e.g., Johnson and Morton 1991). Eyes may be particularly important: people can behave more prosocially when they are watched by even static images of eyes, littering less (Ernest-Jones, Nettle, and Bateson 2011) and voluntarily paying more for coffee with an "honesty box" (Bateson, Nettle, and Roberts 2006).
6. Note that respondents' beliefs about the confidentiality of their answers could go either way: respondents could be more candid with a computer because they believe it will not disclose sensitive information, or they could be more candid with a human because they trust the human's confidentiality assurance more than a computer's (as Aquilino [1994] has argued).

differences is human interviewers' movement and responsivity, not the fact that they are human.[7] We will call this the *Animation* hypothesis.

Our study compares one set—out of many possible—interviewing modes that allowed us to systematically match and vary the channels of facial motion across the different experimental conditions.[8] The advantage of using motion-capture technology (rather than manually animating the virtual interviewers) is that the same human interviewer's voice and facial motion could be used as source material for all the computer interfaces—the same interviewer who administered the FTF interviews. This allowed for fair comparisons across the conditions.

The strategy in the current study was to administer a set of questions that had either led to mode differences (human interviewer versus self-administration) in large samples or that plausibly could lead to such mode differences. The analytic approach was then to compare patterns of response across all four modes (FTF, HA, LA, and ACASI) for only those questions that had demonstrated a FTF versus an ACASI difference. Thus, we could examine whether the mode effects resulted from the interviewer's being human rather than a machine (Human Interviewer hypothesis), the interviewer's motion or responsivity (Animation hypothesis), or the fact that there was a moving face versus no face at all (Facial Representation hypothesis).

Various patterns of results consistent with these hypotheses are possible. The clearest pattern would be a single "dividing line" for all questions that produce a mode effect. For example, if all three automated systems produce the same response distributions and they differ from those produced by the human interviewer, this would make a clear case that the locus of FTF-ACASI mode effects is the presence or absence of a human. Equally straightforward would be a pattern showing a single dividing line elsewhere (e.g., between modes with faces versus ACASI). An alternative possibility is a monotonic increase in socially desirable answers from ACASI to FTF, suggesting that there is an independent contribution of each additional set of features of humanness to respondents' reluctance to disclose. It is also possible that the dividing line is different for different survey items, suggesting that the cause of FTF-ACASI mode differences may vary across questions. Finally, it is possible that more than one hypothesis is supported for any one

7. The impact of animation on users' experience and cognition has been investigated in interface domains other than surveys, including education (e.g., papers in Lowe and Schnotz [2007]) and comprehension of graphs (e.g., Tversky, Morrison, and Betrancourt 2002). A particularly effective use of animation coordinates gesture and speech of embodied animated agents (Cassell 2000; Cassell et al. 2000).

8. One could certainly study the locus of ACASI mode effects through mode comparisons other than those used in this study. For example, one could compare responses from FTF interviews, video-mediated interviews (e.g., via Skype or FaceTime), and video recordings of human interviewers. Or one could compare live human audio interviews (telephone), advanced automated speech dialogue systems, and simple ACASI.

question; this would suggest that multiple causes for FTF-ACASI mode differences can jointly affect a response.

Experiment

DESIGN AND PROCEDURE

Respondents were randomly assigned to one of four experimental conditions. They participated in a forty-two-question interview conducted in the same environment, a university laboratory in New York City, by either 1) a human interviewer (FTF); 2) a High Animation (HA) virtual interviewer; 3) a Low Animation (LA) virtual interviewer; or 4) an Audio Computer-Assisted Self-Interviewing (ACASI) system. (Our implementation of ACASI differed from typical administration in that there was no section of the interview conducted by an FTF interviewer, nor was an interviewer in the room during the self-administered sessions.) In all conditions, a human experimenter (not the interviewer) brought respondents into the lab and oriented them to the interview before it started. This experimenter provided the same confidentiality assurances to all respondents to reduce the possibility that respondents' trust in data confidentiality would differ across modes.

After the interview, the experimenter returned and accompanied respondents to a different room, where they completed a text-based questionnaire in a web browser (not used for computer-administered interviews) that asked respondents to rate how they experienced the interview (e.g., their comfort, their enjoyment, and how natural they found the interaction). In addition, respondents answered questions about their experience with computers in general.

In the FTF condition, the respondent and interviewer sat facing each other. The interviewer read aloud questions presented on the screen of a laptop computer. Respondents answered by speaking, and the interviewer entered the responses into the laptop. In the three computer-based conditions (HA, LA, and ACASI), each respondent sat alone in the same laboratory room used in the FTF condition, facing a desktop computer and wearing headphones. In the HA and LA conditions, the virtual interviewer was displayed on the computer screen above clickable (radio button) response options or a data entry field into which respondents could type a number, depending on the question (see figure 1). There was also a "Repeat Question" button to re-present the question and a "Next" button to advance to the next question. In the ACASI condition, there was no visual display of the virtual interviewer; response options and the "Repeat Question" and "Next" buttons were displayed just as in the HA and LA conditions while respondents heard the same audiorecorded questions presented in the HA and LA interfaces.

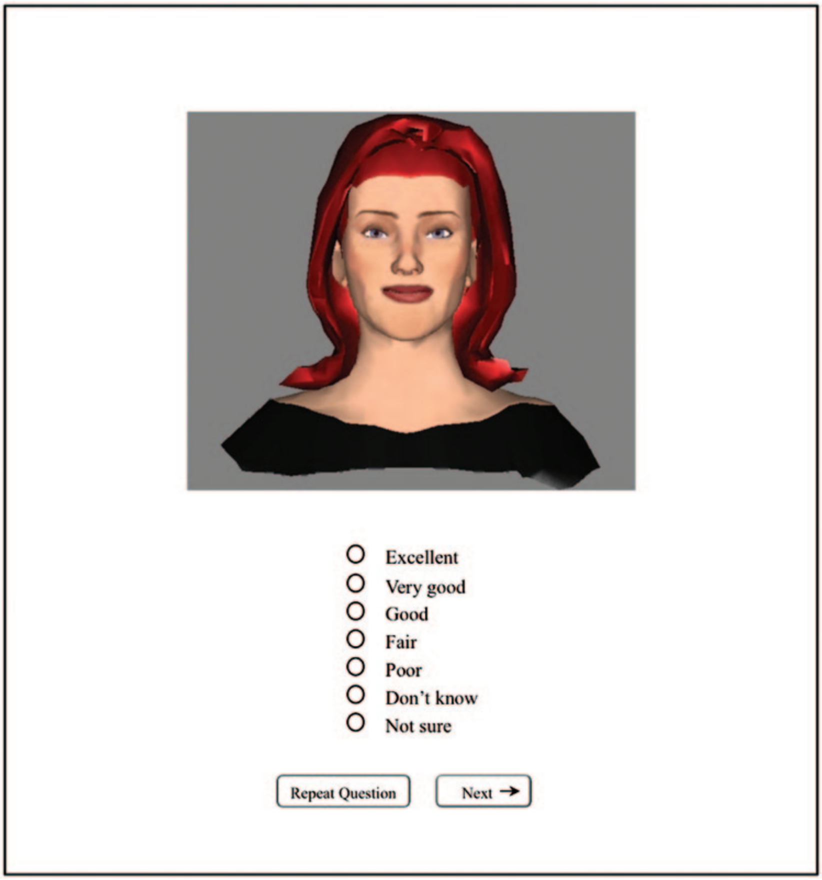

Figure 1. Virtual Interviewer User Interface (with more facial movement in HA than LA condition).

INTERVIEWING INTERFACES

The HA and LA virtual interviewers were created using motion-capture software (proFace Video) for extracting motion from a moving human face.[9] As table 2 lists in detail, the HA and LA virtual interviewers differed in several ways so as to make the HA interviewer more like a human interviewer; see

9. The process involved three main steps. First, blue and green foam dots were affixed to specific places on the human interviewer's face. Second, a head-mounted video camera worn by the interviewer and aimed at her face recorded her face (including the dots) while asking the survey questions chosen for the study. Third, the movement of the dots was extracted from the recorded video via the motion-capture software and applied to a computer-animated head so that the human interviewer's prerecorded facial movements controlled the virtual interviewer's face.

Table 2. Features of Interviewing Agents

	Interviewing agents			
	Face-to-face	High animation	Low animation	ACASI
Audio features				
Speech	Live	Recorded	Recorded	Recorded
Question repeat available	Yes	Yes	Yes	Yes
Question repeated with intro (i.e., "Let me repeat the question…")	Yes	Yes	No	No
Transitions between survey question topics (i.e., "Now I'm going to ask you some questions about personal finances.")	Yes	Yes	Yes	Yes
Visual features				
Facial representation of interviewer	Yes	Yes	Yes	No
Fade-in of video clips	N/A	Yes	No	N/A
Mouth moves in sync with speech	Yes	Yes	Yes	N/A
Head movement	Yes	Yes	No	N/A
Eye blinking	Yes	Yes	No	N/A
Facial expressions	Yes	Yes	No	N/A
Continual evidence of attention (i.e., head movement, eye blinks, facial motion) while respondent is answering question[a]	Yes	Yes	No	N/A
Continual gaze at respondent	No	Yes	Yes	N/A
Interaction				
Respondent answers by speaking	Yes	No	No	No
Respondent able to refuse to answer a question	Yes	Yes	Yes	Yes
Mode of nonresponse prompt	Spoken	Spoken	Textual	Textual
Acknowledgment of respondent actions	Spoken	Spoken	None	None

[a]LA interface displayed same mouth movement as HA, but no other facial movement.

online appendix C for video examples of the HA and LA interviewers (see the supplementary material online). For the HA interviewer, all captured motion was applied to the computer-animated head. For the LA interviewer, only a subset of the captured motion was applied, such that it exhibited no facial expressions, no eye blinking, and no head movement. In the ACASI condition, the interview was presented entirely in audio—the same audio as in the HA and LA conditions.

RESPONDENTS

Two hundred thirty-five respondents from the New York City area were recruited via an online advertisement (on Craigslist New York City) and in print and online ads (in the *Village Voice)*, each of which offered $20 for participating in a survey. The sample consisted of 132 women and 103 men, with a mean age of thirty-one years. The respondents were randomly assigned to one of four experimental conditions, with fifty-nine assigned to the FTF condition, sixty to the HA condition, fifty-seven to the LA condition, and fifty-nine to the ACASI condition. There were no significant differences among the groups in age, race, level of education, or experience with computers.[10]

SURVEY QUESTIONS

Forty-two questions were selected from three well-known and consequential ongoing U.S. social surveys: the Behavioral Risk Factor Surveillance System (BRFSS), the Survey of Consumer Finances (SCF), and the General Social Survey (GSS). The questions concerned seven different topics: health and fitness, sexual health and behaviors, alcohol use, leisure activities, political activism, personal finances, and altruistic behaviors. Of the forty-two questions, twelve were chosen because they had produced mode effects (interviewer administration versus self-administration or FTF versus phone) in large-scale surveys, and twenty-one were chosen either because socially desirable responding had been observed in other studies or because we judged some responses to be more socially desirable than others, and so it was conceivable that mode effects could appear in our smaller lab sample. Nine additional ("neutral") questions for which no mode effects were expected were included to make the survey feel cohesive. (See appendix A for the full list of questions and the reasons for each question's inclusion).

For the twelve questions that had produced mode effects in large samples, we derived predictions about the reporting of socially undesirable behaviors.

10. Experience with computers was measured in several ways: self-reported time spent reading and sending email per week; time spent on the web for other tasks; time spent in chat rooms, newsgroups, or on bulletin boards; use of Facebook and other social media; experience with avatars in online settings; self-reported Internet expertise; and a combined digital literacy measure (Hargittai 2005) based on questions asked in the Internet Modules of the General Social Survey in 2000 and 2003.

On ten of the questions, respondents in the published studies reported more socially undesirable or embarrassing behaviors in the more anonymous mode, and so we would expect respondents to report more of those behaviors in our ACASI condition than FTF. For two questions (smoking and weight), the previously demonstrated mode effects (Béland and St-Pierre 2008) were in the opposite direction than has been typically observed, with *more* reporting of sensitive information FTF than in telephone interviews. If we assume that telephone interviews afford respondents more privacy and distance from the interviewer than FTF, we should see a similar reversal in the patterns of mode differences here, with more reports of smoking and being overweight to FTF interviewers than to ACASI. For the twenty-one questions that we deemed potentially likely to produce mode differences in socially desirable responding, we would expect more candid (less socially desirable) responding in ACASI than in FTF. For the nine "neutral" questions, we would expect no differences in responding FTF versus ACASI.

STATISTICAL METHODS

We used ANOVA to test mode differences in the twelve questions requiring open numerical (continuous) responses; logistic regression for the eight questions that required yes/no responses; ordinal logistic regression for the nineteen questions requiring selection from an ordered response scale (e.g., *Every day*, *Several times a week*, *Several times a month*, *Rarely*, or *Never*); and multinomial logistic regression for the three questions requiring a choice from an unordered response scale. For six numerical questions (male sex partners, female sex partners, sex partners in last year, sex partners since age eighteen,[11] new credit card charges, and drinking regularly), the response distribution was skewed to the right; for these questions, we log-transformed the data and examined not only the full distribution of responses, but also the response patterns for the top quartile of responses. Focusing on respondents whose reports are largest makes sense on theoretical grounds: a respondent who never drinks or has never had a sex partner, for example, cannot report less drinking or fewer sex partners, while a respondent who drinks frequently or has had many sex partners has more "room" to underreport the total with a human interviewer or report a higher, more accurate total in ACASI. And presumably those who have more to report experience more social pressure to misreport, although of course norms may vary in different subpopulations. So mode effects that may not emerge in a distribution with many zero values may be visible among the subgroup with the most to report.

11. This is a composite created by summing reported male and female partners since age eighteen.

Results

QUESTIONS THAT PRODUCED FTF VERSUS ACASI MODE EFFECTS

Recall that our analytic strategy was to compare patterns of response across all four modes only for those questions whose answers differed FTF versus ACASI. For the nine questions for which we did not expect mode differences, there were no reliable differences between these modes. For the thirty-three questions for which mode effects seemed plausible, answers for ten of them differed significantly or marginally between FTF and ACASI, including one (sex partners since age eighteen) for which the effect was found for the top quartile of responses but not the full distribution; these are the questions on which we will focus. (It is not apparent from the content of the questions why these items produced the effect and the others did not.[12]) For nine of these ten questions, the FTF versus ACASI difference was in the expected direction based on previously published mode effects or findings on socially desirable responding. (For eight of these nine, we expected more socially desirable responses FTF than ACASI; and for one, smoking, we expected the opposite.) For the remaining question (drinking regularly), the difference was in the unexpected direction: more regular drinking was reported in FTF than ACASI. The leftmost columns of table 3 (FTF versus ACASI) summarize these effects.

RESPONSE DISTRIBUTIONS

Table 4 presents the summary measure (mean, median, or percentage) for each of these ten questions that, in our judgment, is most revealing about the patterns of responding. For example, the question about watching primetime television offers the following ordered response options: “every day,” “several times a week,” “several times a month,” “rarely,” and “never.” For this question, table 4 presents the percentage of respondents who reported watching primetime television several times a week or more. (See appendix B for the exhaustive response distributions.)

Where did responses to the virtual interviewers fall relative to FTF and ACASI? As table 4 and appendix B show, for all ten questions responses to both virtual interviewers fall in between the FTF and ACASI responses.

TESTING THE THREE HYPOTHESES

We carried out a series of comparisons between our experimental conditions (see table 3) to test our hypotheses. To test the Human Interviewer hypothesis,

12. Note that our observing FTF-ACASI differences for ten of the thirty-three sensitive questions in the survey is consistent with previously reported studies, where mode differences are most often found for a subset of sensitive questions in a survey—sometimes for only one or two.

Table 3. Mode Comparisons for the Ten Questions Showing a Significant FTF-ACASI Effect

Model type	Survey question	Face to face vs. ACASI		Human vs. computer (Human Interviewer hypothesis)		High animation vs. low animation (Animation hypothesis)		Faces vs. ACASI (Facial Representation hypothesis)	
		B (SE)	Partial eta squared	B (SE)	Partial eta squared	B (SE)	Partial eta squared	B (SE)	Partial eta squared
ANOVA	SexPartners18±	–0.75* (0.35)	0.08	–0.36 (0.29)	0.03	–0.48# (0.25)	0.06	–0.66* (0.29)	0.09
	TVHours	–0.61# (0.36)	0.01	–0.52# (0.29)	0.01	–0.17 (0.26)	0.002	–0.30 (0.30)	0.004
Model type	Survey question	B (SE)	Odds ratio	B (SE)	Odds ratio	B (SE)	Odds ratio	B (SE)	Odds ratio
Logistic regression	Smoke	1.06** (0.40)	2.90	0.48 (0.30)	1.62	0.60* (0.27)	1.82	0.92* (0.34)	2.52
	HaveCreditCards	0.89* (0.41)	2.43	0.55 (0.35)	1.74	0.24 (0.28)	1.27	0.63* (0.32)	1.88

Continued

Table 3. *Continued*

Model type	Survey question	Face to face vs. ACASI		Human vs. computer (Human Interviewer hypothesis)		High animation vs. low animation (Animation hypothesis)		Faces vs. ACASI (Facial Representation hypothesis)	
		B (SE)	Partial eta squared	B (SE)	Partial eta squared	B (SE)	Partial eta squared	B (SE)	Partial eta squared
Ordinal logistic regression°	RegularlyDrink	0.62# (0.33)	1.86	0.43 (0.27)	1.54	0.09 (0.23)	1.09	0.41 (0.26)	1.51
	Newspaper	0.94** (0.34)	2.56	0.64* (0.28)	1.90	0.48* (0.24)	1.62	0.62* (0.27)	1.86
	TVPrimetime	–0.73* (0.33)	0.48	–0.45# (0.27)	0.64	–0.25 (0.23)	0.78	–0.53# (0.27)	0.59
	Savings	0.64# (0.34)	1.90	0.39 (0.28)	1.48	0.32 (0.24)	1.38	0.47# (0.28)	1.60
	GivingHomeless°	0.60# (0.33)	1.82	0.33 (0.28)	1.39	0.17 (0.24)	1.19	0.34* (0.16)	N/A°
	SeatStranger	0.72* (0.33)	2.05	0.47# (0.27)	1.60	0.21 (0.24)	1.23	0.48# (0.27)	1.62

NOTE.—Models are presented by the type of response: ANOVA models for numerical (continuous) responses, logistic regression models for yes/no responses, and ordinal logistic regression models for ordered response scales. Parameter estimates (B) are given with standard errors in parentheses (SE). Effect sizes are also presented.

°Ordinal logistic models were fit using the logit link except in the case where the parallel lines assumption failed: Faces vs. ACASI estimate for GivingHomeless question based on probit link; coefficients for logit link models reported on log-odds scale.

± For upper quartile of respondents (number of sex partners >= 20), coefficients reported on the log scale.

#$p \leq 0.10$; *$p \leq .05$; **$p \leq .01$

Table 4. Summarized Patterns of Responding

	Face-to-face	High animation	Low animation	Audio computer-assisted self-interview
Hours watching TV per day (mean and SE)	2.2 (0.24)	2.8 (0.24)	2.5 (0.28)	2.8 (0.27)
Number of sex partners since 18th birthday[a] (median, *N*)	41.5 (16)	45 (11)	47 (16)	80 (14)
Have credit cards (*N*, %)	46 (78.0)	40 (66.7)	43 (75.4)	35 (59.3)
Smoked 100 cigarettes in entire life (*N*, %)	30 (50.8)	27 (45.0)	23 (40.4)	15 (25.4)
Reading newspaper every day (*N*, %)	29 (49.2)	23 (38.3)	20 (35.1)	11 (18.6)
Watching primetime TV several times a week or more (*N*, %)	22 (37.3)	30 (50.0)	29 (50.9)	36 (61.0)
Save regularly each month (*N*, %)	17 (28.8)	17 (28.3)	12 (21.1)	8 (13.6)
Drink 3 times a week or more (N, %)	18 (30.5)	15 (25.0)	16 (28.1)	5 (8.5)
Giving seat to a stranger at least once a week (*N*, %)	21 (35.6)	17 (30.4)	19 (34.0)	12 (21.0)
Giving food or money to a homeless person at least once a week (*N*, %)	10 (17.0)	9 (15.0)	9 (15.8)	1 (1.7)

[a]Top quartile.

we compared responses FTF to all three computer-administered conditions. To test the Animation hypothesis, we compared FTF and HA responses to LA and ACASI responses. And to test the Facial Representation hypothesis, we compared ACASI responses to responses in all three conditions in which a face was represented (FTF, HA, and LA conditions).[13]

As shown in table 3, the Human Interviewer hypothesis was supported significantly in one question (newspaper reading) of the ten, and marginally supported in two more (TV hours watched and giving a seat to a stranger). For one of these questions (TV hours), the Human Interviewer hypothesis was the only hypothesis supported (marginally); apparently, for this question it is the presence of a living human interviewer—something about the difference between FTF and HA interviewers—that drives the FTF versus ACASI effect.

The Animation hypothesis was supported in two questions (smoking and newspaper reading) and marginally supported in one more (sex partners since age eighteen), although it was not the only hypothesis supported for these questions. The responses for HA and LA virtual interviewers (table 4) seem more similar on these questions than one might expect if the amount of animation is the locus of the FTF-ACASI effects. To examine this, we carried out two additional comparisons between 1) responses to the two virtual interviewers (High and Low Animation); and 2) responses to the two virtual interviewers and FTF. There were no reliable differences for any of the questions in either comparison. So, the support for the Animation hypothesis in these data may be a statistical artifact rather than an explanation: if amount of animation is responsible for the FTF-ACASI difference for these questions, then surely responses to HA and LA should differ and responses to the automated systems should differ from FTF, but they do not.

The Facial Representation hypothesis, in contrast, is supported for eight of the ten questions that had shown an FTF versus ACASI difference; it is supported significantly in five of the ten questions, and marginally supported in three more. The Facial Representation hypothesis is the only hypothesis supported for three of these eight questions (Credit cards, Savings, and Giving to the homeless), which suggests that the presence of a face is indeed the locus of the FTF-ACASI effect for these questions. For the other five questions, other hypotheses are also supported: Animation for three and Human Interviewer for three (both Animation and Human Interviewer hypotheses are supported for the sex partners question).

13. See online appendix D for all comparisons—FTF versus ACASI and the hypothesis tests, as well as two additional comparisons (ACASI versus HA and LA, and HA versus LA)—for all forty-two questions in the survey. As the appendix shows, there were twelve marginally significant ($p < .10$) and four significant ($p < 0.05$) effects out of the 242 additional comparisons beyond those presented in table 4. This is no more than would be expected by chance for this many comparisons and suggests that our analytic strategy of testing our hypotheses on the ten questions for which FTF-ACASI mode differences were observed is appropriate.

Underscoring the notion that all modes with faces tend to group together are the additional analyses indicating no differences in responses to the two virtual interviewers and FTF. Because of this, the Animation hypothesis seems less viable as an explanation in the cases where the Facial Representation hypothesis is also supported. The locus of the FTF-ACASI mode difference for these questions (three for which only the Facial Representation hypothesis was significantly supported and two for which both Animation and Facial Representation hypotheses were supported) thus seems to be the presence versus the absence of a face (live or virtual).

The one question for which the FTF versus ACASI difference was not in the expected direction (drinking regularly) did not yield support for any of the three hypotheses. There were no other reliable differences between the conditions in any other comparisons, although for the top quartile of responses there were several effects we could not interpret; we do not report them here because there were no FTF-ACASI differences among these responses. We do not have a direct explanation for either the reversal or the lack of support for any of the hypotheses, but note that the frequency of regular drinking in this sample was quite low. It may be that this question was therefore not as sensitive for these respondents as has been seen in previously reported findings, and that the difference does not reflect socially desirable responding. Given this, the reversal of the FTF versus ACASI mode effect may not allow real testing of our hypotheses for this question in this sample.

Overall, the patterns demonstrated different "dividing lines" (discontinuities in disclosure between different groupings of modes) across the ten questions, with more than one hypothesis supported for some questions. This clearly argues against an account of FTF-ACASI mode differences based only on whether a live human interviewer asks the questions and records the answers (the Human Interviewer hypothesis). The hypothesis receiving the most support (in eight of the ten questions, uniquely in three) was the Facial Representation hypothesis, suggesting that even a clearly nonhuman facial representation with limited motion can reduce disclosure as much as an actual human interviewer relative to a voice alone.

More generally, the pattern is consistent with our suggestion that human interviewing comprises a bundle of features which, when deployed independently in survey interviewing systems, can independently affect disclosure. What is less clear is why certain combinations of features affected answers for the particular questions that they did—e.g., support for the Human Interviewer hypothesis for TV hours watched, but support for the Facial Representation hypothesis for having a credit card.

We don't see anything in the content of these questions that would lead to one pattern of results rather than another, although surely the content matters. In fact, we suspect that there may not be a simple mapping between domain of questioning and the kinds of mode effects we are documenting here. Can the patterns instead be explained by how respondents experienced the different interviewing modes? Anecdotally, we observed a wide range of reactions to the virtual interviewers in the optional postsurvey comments, from the less positive (e.g., "I was a little distracted by her moving lips. A little creepy") to the more enthusiastic ("I thought she looked great, like real people do—good teeth smile [*sic*], nice hair, very realistic, and easy to follow which I enjoyed"). If respondents' willingness to disclose is affected by how embarrassed or uncomfortable they feel interacting with an interviewer or interviewing agent (along the lines proposed by, e.g., Schaeffer [2000]; Tourangeau and Yan [2007]), one might predict that respondents interviewed in the modes with faces, for example, would have reported feeling less comfortable with the interview than those interviewed via ACASI. To investigate this further, we examined the postinterview questionnaire responses more closely.

POSTINTERVIEW RATINGS

We had complete postinterview questionnaires for 213 of the 235 respondents: fifty-seven FTF, fifty-eight HA, fifty-one LA, and forty-seven ACASI. Prior analyses that had shown FTF-ACASI differences in the full sample were repeated on this subset and resulted in similar patterns of mode differences. We interpreted this as indicating sufficient similarity between the subsample and full sample to justify further analyses.

To the extent that there were differences in ratings across modes, FTF and ACASI respondents rated their overall experience of the interview more positively than either HA or LA respondents. Respondents rated their interaction as more *natural* (on a five-point scale) in both FTF and ACASI than HA and LA (for FTF compared to HA and LA, $F[2,203] = 10.44$, $p < 0.001$; for ACASI compared to HA and LA, $F[2,203] = 3.74$, $p = 0.03$). They also rated the ACASI interviewer as more *like a human* (as opposed to more like a computer) on a three-point scale than the HA and LA interviewers ($F[2,148] = 5.31$, $p = 0.01$); the HA and LA interviewers were rated equally human, ($F[1,148] = 0.63$, *n.s.*).

Respondents rated the FTF interviewer more positively than the computer-based conditions. Respondents reported that their *comfort* level increased over the course of the interview more (on a three-point scale) in FTF ($F[1,211] = 5.25$, $p = 0.02$); that they *enjoyed* the FTF interview more (on a five-point scale) than the computer-based interviews ($F[1,212] = 9.53$, $p < 0.01$); and that they found the FTF interview less *frustrating* (on a five-point scale) than the computer-based interviews ($F[1,210] = 10.56$, $p = 0.001$).

These patterns of ratings do not seem consistent with the disclosure findings. The fact that respondents' ratings of the interviewers' naturalness in FTF and ACASI did not differ is inconsistent with our observed FTF-ACASI disclosure difference and with any patterns of disclosure that we observed across the four modes. The ratings of comfort, enjoyment, and frustration are in one sense consistent with the Human Interviewer hypothesis in that they differentiate between the human interviewer and the computer-based modes, but they seem at odds with the findings on disclosure; one might expect respondents who disclose less to report greater discomfort, less enjoyment, and more frustration in FTF than in the computer-based modes. Perhaps respondents who disclosed less feel happier about the interview.

To further clarify the relationship between the disclosure patterns and respondents' reported feelings about the interview, we looked more closely at differences among respondents within the experimental groups. We dichotomized respondents into those who reported being very or extremely comfortable at the start of the interaction and those who did not, as well as into those who reported enjoying the interview somewhat or thoroughly and those who did not, and those who rated their interaction as more natural and less natural. (Of course, these variables do not measure precisely the same underlying construct, but they all can be argued to reflect comfort or a sense of human connection.) We then tested whether the FTF-ACASI difference in disclosure (for the ten questions that had shown a difference) interacted with any of these three dichotomized affect measures.

Figure 2 shows the pattern that we observed for the question about giving a seat to a stranger. Respondents who reported having been more comfortable at the start of the interview showed a typical FTF-ACASI mode effect on disclosure: respondents were more likely to report that they had not given a seat to a stranger in the past year in ACASI than FTF. Respondents who reported not having been comfortable showed no effect of mode, interaction of comfort (yes-no) and mode (FTF-ACASI), $Z = -2.90$, $p = .004$. Of course, we do not know the direction of causation: on the one hand, respondents who felt comfortable with the FTF interviewer could as a result have been more concerned about how the interviewer perceived them and so disclosed less than respondents who felt uncomfortable with the interviewer (e.g., less connected, less rapport) and thus did not mind disclosing the socially undesirable behavior. On the other hand (reversing the causality), respondents who had little to disclose (or felt unembarrassed about not giving a seat to a stranger) may have felt more positive about the interaction, while respondents with more to disclose may have felt worse. The same causal ambiguity holds for ACASI: respondents who felt comfortable with the ACASI system may have felt uninhibited and thus disclosed more, or respondents who had much to disclose may have felt more positive about their disclosure when made to an automated system.

Either way, this finding suggests that how respondents experience the interview is related to how much they disclose, and that ACASI-FTF mode effects may be driven by a subset of the respondents. We observed the same pattern for three more of the ten survey questions (two significantly and one marginally) on at least one dimension of reported experience (naturalness, comfort, humanness). Clearly, these data are more suggestive than definitive, but the patterns are consistent enough to justify further investigating an experience-based account of the FTF-ACASI mode effect.

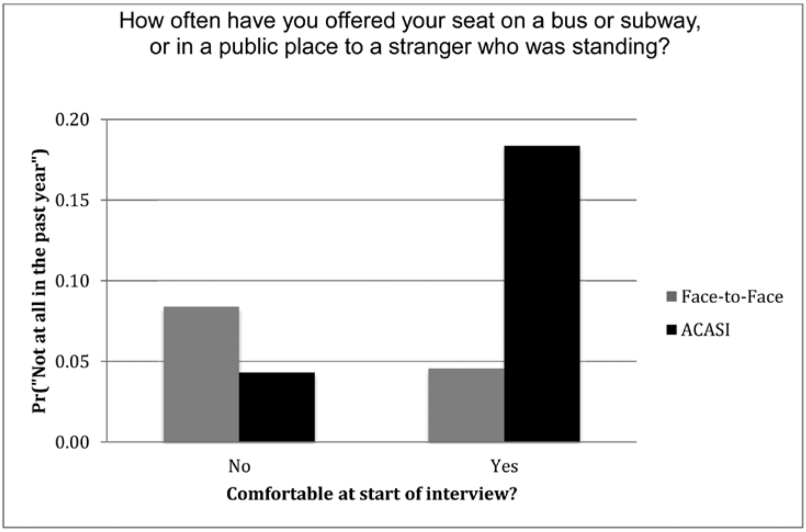

Figure 2. Marginal Probability of Giving the Least Socially Desirable Response ("Not at all in the past year") for Respondents Who Reported Being Comfortable vs. Not Comfortable at the Start of the Interview.

Discussion and Implications

It has long been known that interviewer administration of survey questions (compared to self-administration) can suppress disclosure and increase socially desirable answers, presumably because respondents wish to avoid embarrassing interviewer reactions and other potentially harmful consequences, such as disclosure to family members or law enforcement. The current study further explores what it is about the presence of a human interviewer that is responsible for survey mode effects of this type, both to better understand the origins of these effects and to potentially predict when survey modes, whether current or future, are likely to affect answers. The proposal is that different aspects of human interviewers—embodiment, perceptual capability, dialogue capability, voice, etc.—can *independently* affect respondents' willingness to disclose or to misreport, and thus that FTF-ACASI mode effects do not result simply from the presence or absence of a human interviewer. Instead, the particular combination of these attributes in a survey mode leads to a particular level of disclosure.

Our findings largely support this proposal. Specifically, they demonstrate that embodiment in the form of a face even with limited motion can increase socially desirable responding (consistent with Sproull et al. [1996]) just as much whether the face is biological or digital—the Facial Representation hypothesis—and even in a relatively small sample. Even though the presence of a human *can* drive the mode effects (as it did uniquely for our survey question about TV hours watched), disclosure can be inhibited by interfaces that have only some features of a human interviewer. Beyond this, the findings rule out certain possible explanations for the causes of FTF-ACASI mode effects. For example, support for the Facial Representation hypothesis cannot result from an unnaturally fixed gaze by the interviewers: HA and LA interviewers gazed fixedly at respondents, but the FTF interviewer's gaze varied because she needed to look at her laptop; nonetheless, responses in all three modes grouped together for eight questions. Similarly, support for the Facial Representation hypothesis cannot be attributed to how respondents registered their answers. In all three automated modes (HA, LA, and ACASI), respondents entered their answers textually and with the mouse while FTF respondents spoke their answers. Yet, the automated modes with faces grouped with FTF and not ACASI.

Our results extend the domains of questioning for which FTF-ACASI mode effects have been observed. For example, prior studies on mode effects in answering questions about smoking have observed more reported smoking FTF than on the telephone (Gunderson, Delnevo, and Momperousse 2007) or more reported smoking in paper-and-pencil self-administration than FTF (Brittingham, Tourangeau, and Kay 1998); here, we see an FTF-ACASI mode effect with greater reports of smoking not only FTF but in the self-administered modes with a face (HA and LA). This is curious if the explanation for the mode effect is that respondents feel they can't lie about smoking to a live interviewer because the interviewer can smell smoke and see tobacco stains; our virtual interviewers had no such perceptual capability. Perhaps embodiment—digital or human—evokes a sense that the interlocutor has perceptual capabilities that of course on closer inspection would be dismissed. Alternatively (and equally curious), in the domain of smoking, respondents may have experienced a level of rapport with a digital, moving face equivalent to their rapport with a live interviewer, leading to similar increases in disclosure.

Of course, we do not know which reports by respondents are true; the fact that less socially desirable answers in other domains tend to be more accurate (Kreuter, Presser, and Tourangeau 2008) increases the plausibility of interpreting our results as we do, but there is obviously no guarantee. Nonetheless, the findings are compelling given that they emerged in a relatively small convenience sample, i.e., with limited statistical power, and the patterns of data are generally as one would expect given prior evidence (with the exception of drinking regularly). The experimental design also reduces the plausibility of alternative explanations for the pattern of results; it is unlikely that the results are due to differences in respondents' motivations to participate (respondents were randomly assigned to conditions, and they all received the same monetary incentive), different response rates across the conditions (respondents did not know which condition they would be assigned to when they made the decision to participate and visit the lab), or different evidence about the likely confidentiality of the data (respondents were told by a human experimenter before the survey that the data would be kept confidential in the same way across all four conditions).

The findings reported here add to an emerging (though still incomplete) picture about the particular combinations of features in an interviewing interface that reduce disclosure. Thus far, the addition of still faces to a web interface does not seem to affect disclosure of sensitive information (Tourangeau, Couper, and Steiger 2003). Recorded moving faces, however, do seem to reduce disclosure, at least when accompanied by a voice, compared to disclosure through a textual interface; for two additional examples, see Sproull et al. (1996), who found higher social desirability scores and reports of greater altruism to talking heads than to text, and Fuchs (2009), who found reduced disclosure of sexually transmitted diseases and certain sexual behaviors, relative to text, when a recorded video interviewer's gender differed from the respondent's, but in some cases greater disclosure to a same-gender recorded video interviewer. This starts to suggest that there is something special about moving, speaking faces—but they don't have to be human, live (as opposed to recorded), of particularly high fidelity, or particularly reactive to respondents' behaviors. From the other end, it suggests that what is special about ACASI is that the recorded speech is disembodied—it lacks a moving face.

The proposal that interviewing modes are collections of features raises the possibility of designing automated interviewing systems that strategically tailor particular combinations of features to maximize respondents' willingness to disclose sensitive information. One might implement the same set of features for all respondents, different features for different subgroups of respondents, different features for individual respondents, and even different features for different topic domains or questions within an interview. Potential benefits include increasing (some) respondents' engagement, effort, and completion rates by including particular features (like a moving face) for some respondents (e.g., the kind of respondent who reported enthusiasm for our virtual interviewers) or for some questions (e.g., the smoking question or "neutral" questions). Potential drawbacks include reduced disclosure, disengagement, or increased breakoffs for other respondents (e.g., the kind who found the virtual interviewer creepy) or other questions (e.g., altruistic behaviors).

To implement this, ideally one would know, for each respondent in a sample and for every question, several key attributes. First is *the respondent's general propensity to disclose*. Some respondents may be predisposed to disclose in any mode, or about particular topics in any mode, and others may avoid disclosure in general or about a particular topic in any mode. Respondents in the middle on this disclosure propensity continuum should be most likely to be affected by features of an interviewing mode.

A second attribute useful to know would be *the likelihood for a respondent that the question is sensitive*. Respondents who have more to hide may feel particularly embarrassed or threatened with a more humanlike interviewing interface, and more comfortable when human features are minimal. Respondents who feel they have nothing to hide are less likely to be affected by interview mode. Some topics are likely to be sensitive for everyone (e.g., income questions in the United States and Europe), while questions on other topics may be sensitive only for subgroups (e.g., heroin use is unlikely to be sensitive for nonusers) (see Schaeffer 2000).

Another potentially useful attribute would be *respondents' affective reactions to an interface*. Respondents may vary in how much rapport or embarrassment they feel interacting with a human interviewer or with a humanlike interface.

Knowing how these attributes are distributed across the sample for a particular question could allow one to predict whether mode effects are likely, and thus whether an adapted interface might be effective. In our view, mode effects should depend on the proportion of respondents who are affected by mode and the size of the mode difference for those who are affected, just as nonresponse error depends on both response rate and the size of the difference in what is being measured between respondents and nonrespondents (e.g., Groves et al. 2009). From a practical perspective, some of this information is inferable from published evidence about general trends in the population and, potentially, from data about individual respondents collected in prior waves of a panel study or earlier in a single interview.

The current study does not compare all possible humanizing features that could affect disclosure. Our virtual interviewers involved one head paired with one voice with movement of one human actor; whether the effects would be the same with a different human actor or different interface characteristics needs to be verified. Although the effects emerged across several topic domains, it is also unknown when and how they extend to other domains. And the fact that multiple features differed across the experimental conditions does not allow conclusions about the relative contributions to disclosure of those different features. Nonetheless, the findings demonstrate that the well-known differences in responses to human interviewers and self-administered computer systems do not result simply from the fact the computers aren't human; rather, individual or bundled features of humanness in a user interface—especially a facial representation—can still reduce disclosure.

Appendix A.

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Would you say that in general your health is excellent, very good, good, fair, poor, don't know, or you're not sure? (GeneralHealth)	Behavioral Risk Factor Surveillance System (BRFSS)			X	Ordered response categories				
A routine checkup is a general physical exam, not an exam for a specific injury, illness, or condition. How long has it been since you last visited a doctor or other health care provider for a routine checkup? Never, 5 or more years ago, within the past 5 years, within the past 2 years, within the past year, within the past 6 months, or not sure? (SeeDoctor)	BRFSS			X	Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
How do you describe your weight? Very underweight, slightly underweight, about the right weight, slightly over-weight, or very overweight? (Weight)	BRFSS	X			Ordered response categories	Béland and St-Pierre 2008	CAPI/ CATI	Lower reports of obe-sity in CATI	
On how many of the past 7 days did you exercise or participate in physical activity for at least 20 minutes that made you sweat and breathe hard, such as basketball, soc-cer, running, swimming laps, fast bicycling, fast dancing, or similar aerobic activities? (DaysExercise)	BRFSS	X			Numeric	Béland and St-Pierre 2008 (Tourangeau et al. 1997 had a very similar finding with a slightly different question in their bogus pipeline experiment)	CAPI/ CATI	Significantly lower reports of exercise in CAPI	

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Thinking about nutrition, how many total servings of fruit and/or vegetables did you eat yesterday? A serving would equal one medium apple, a handful of broccoli, or a cup of carrots. (FruitVegServings)	BRFSS			X	Numeric				
Have you smoked at least 100 cigarettes in your entire life? (Smoke)	BRFSS	X			Yes/no	Béland and St-Pierre 2008; Gundersen, Delnevo & Momperousse 2007	CAPI/ CATI	Higher reports of smoking in FTF mode	
Have you ever been tested for HIV? Do not count tests you may have had as part of a blood donation. (HIVTest)	General Social Survey (GSS)		X		Yes/no				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
How many sex partners have you had in the last 12 months? (SexPartnersYear)	GSS	X			Numeric	Tourangeau et al. 1997; Tourangeau & Smith 1996	FTF/ SAQ	Higher reports for men in FTF; higher reports for women in SAQ	Log transformed
During the past 12 months, have your sex partners been exclusively male, exclusively female, or both male and female? (MFSexPartners)	GSS		X		Unordered response categories				Four unordered categories: none, male only, female only, both male and female
About how often did you have sex during the last 12 months? Not at all, once or twice, once a month, 2–3 times a month, weekly, 2–3 times per week, 4 or more times per week? (HowOftenSex)	GSS		X		Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Now thinking about the time since your 18th birthday, how many female partners have you had sex with? (FemalePartners18)	GSS	X			Numeric	Tourangeau et al. 1997; Tourangeau & Smith 1996	FTF/ SAQ	Higher female reports in SAQ than in FTF	Log transformed
Now thinking about the time since your 18th birthday, how many male partners have you had sex with? (MalePartners18)	GSS	X			Numeric	Tourangeau et al. 1997; Tourangeau & Smith 1996	FTF/ SAQ	Higher female reports in SAQ than in FTF	Log transformed
Now I'm going to read a list of terms that people sometimes use to describe themselves: heterosexual or straight; homosexual, gay, or lesbian; and bisexual. Which term best describes how you think of yourself? (SexualIdentity)	GSS		X		Unordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
A drink of alcohol is one can or bottle of beer, one glass of wine, one can or bottle of wine cooler, one cocktail, or one shot of liquor. During the past 30 days, how many days per week (or per month) did you have at least one drink of any alcoholic beverage? (DaysDrinkPast30Days)	BRFSS	X			Numeric	Aquilino & LoSciuto 1990	SAQ/ CATI	Higher reports of drinking in SAQ mode	
Considering all types of alcoholic beverages, how many times during the past 30 days did you have more than 5 drinks on one occasion? (MoreThan5Drinks)	BRFSS	X			Numeric	Aquilino & LoSciuto 1990	SAQ/ CATI	Higher reports of drinking in SAQ mode	

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Thinking back over the last 12 months, about how regularly did you drink alcoholic beverages? Never in 12 months, 1–3 times in 12 months, 4–7 times in 12 months, 8–11 times in 12 months, 1–3 times a month, once or twice a week, 3–4 times per week, 5 times a week or more? (RegularlyDrink)	BRFSS	X			Ordered response categories	Aquilino & LoSciuto 1990	SAQ/ CATI	Higher reports of drinking in SAQ mode	
Again, as you think back over the last 12 months, how many drinks would you have on a typical day when you drank? (TypicallyDrink)	GSS	X			Numeric	Aquilino & LoSciuto 1990	SAQ/ CATI	Higher reports of drinking in SAQ mode	Log transformed
During the past 12 months, have you read novels, short stories, poems, or plays, other than those required by work or school? (ReadNovels)	GSS			X	Yes/no				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
How often do you read the newspaper—every day, a few times a week, once a week, less than once a week, or never? (Newspaper)	GSS		X		Ordered response categories				
On the average day, about how many hours do you personally watch television? (TVHours)	GSS	X			Numeric	Bridges 2000	Mail/ CATI	Lower reports of television watching in CATI	
I'll now ask about some different kinds of television shows. Would you tell me how often you watch primetime drama or situation comedy programs? Would you say every day, several times a week, several times a month, rarely, or never? (TVPrimetime)	GSS		X		Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
How often do you watch world or national news programs? Every day, several times a week, several times a month, rarely, or never? (TVNews)	GSS		X		Ordered response categories				
And how often do you watch programs shown on public television? Every day, several times a week, several times a month, rarely, or never? (TVPublic)	GSS		X		Ordered response categories				
Which of the following statements best describes you current employment status? Employed for wages, self-employed, out of work for more than one year, out of work for less than one year, a homemaker, a student, retired, unable to work? (EmploymentStatus)	Survey of Consumer Finances (SCF)			X	Unordered response categories				Collapsed to three unordered categories: employed, voluntarily unemployed, involuntarily unemployed

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Now I have some questions about credit cards and charge cards. Do you have any credit cards or charge cards? Please do not include debit cards. (HaveCreditCards)	SCF		X		Yes/no				
Are any of the cards you have any type of Visa, Master Card, Discover, or American Express cards you can pay off over time? (Do not include regular American Express charge cards that must be paid in full.) (CredictCardsPayOverTime)	SCF			X	Yes/no				
How many do you have? Please do not count duplicate cards for the same account or any business or company accounts. (HowManyCreditCards)	SCF		X		Numeric				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
On your last bills, roughly how much were the new charges made to these accounts? (NewChargesCardholders)	SCF		X		Numeric				Log-transformed values reported for cardholders
After the last payments were made on these accounts, roughly what was the balance still owed on these accounts? (Balance)	SCF		X		Numeric				
Thinking only about Visa, Master Card, Discover, American Express cards you can pay off over time, and store cards, do you almost always, sometimes, or hardly ever pay off the total balance owed on the account each month? (CreditCardsPayoffTotal)	SCF		X		Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Now I'd like to ask you some questions about your attitudes about savings. People have different reasons for saving, even though they may not be saving all the time. Which of the following statements comes closest to describing your saving habits? Don't save—usually spend more than income; don't save—usually spend about as much as income; save whatever is left over at the end of the month—no regular plan; save income of one family member, spend the other; spend regular income, save occasional other income; save regularly by putting money aside each month? (Savings)	SCF		X		Ordered response categories				Collapsed to three ordered categories: don't save, save occasionally, save regularly

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
Over the past year, would you say that your spending exceeded your income, that it was about the same as your income, or that you spent less than your income? (Spending)	SCF		X		Ordered response categories				
Some people seem to follow what's going on in government and public affairs most of the time, whether there's an election going on or not. Others aren't that interested. Would you say you follow what's going on in government and public affairs most of the time, some of the time, only now and then, or hardly at all? (FollowPolitics)	SCF			X	Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
In 2004, you may remember that Kerry ran for President on the Democratic ticket against Bush for the Republicans. Did you vote in that election? (Vote2004)	GSS		X		Yes/no				
What about local elections—do you always vote in those, do you sometimes miss one, do you rarely vote, or do you never vote? (VoteLocal)	GSS		X		Ordered response categories				
In the past three or four years, have you attended any political meetings or rallies? (PoliticalRally)	GSS			X	Yes/no				
In the past three or four years, have you contributed money to a political party or candidate or to any other political cause? (PoliticsMoney)	GSS			X	Yes/no				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
During the past 12 months, how often have you donated blood? Not at all in the past year, once a month, two or three times a year, once in the past year, or don't know? (DonateBlood)	GSS		X		Ordered response categories				
During the past 12 months, how often have you given food or money to a homeless person? Not at all in the past year, once in the past year, two or three times a year, once a month, once a week, more than once a week, or don't know? (GivingHomeless)	GSS		X		Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
During the past 12 months, how often have you done volunteer work for a charity? Not at all in the past year, once in the past year, two or three times a year, once a month, once a week, more than once a week, or don't know? (VolunteerWork)	GSS	X			Ordered response categories	Acree et al. 1999	Mail/ CATI	Higher reports of volunteer work in CATI	
During the past 12 months, how often have you given money to a charity? Not at all in the past year, once in the past year, two or three times a year, once a month, once a week, more than once a week, or don't know? (MoneyCharity)	GSS		X		Ordered response categories				

Continued

Appendix A. *Continued*

Question	Survey of origin	Published mode effect	Hypothesized mode effect	No mode effects expected	Dependent variable	Mode effect published	Modes	Direction	Model notes
During the past 12 months, how often have you offered your seat on a bus or in a public place to a stranger who was standing? Not at all in the past year, once in the past year, two or three times a year, once a month, once a week, more than once a week, or don't know? (SeatStranger)	GSS		X		Ordered response categories				

Appendix B. Complete Response Distributions for Ten Questions Showing FTF-ACASI Mode Differences

Question items	FTF $N = 59$	HA $N = 60$	LA $N = 57$	ACASI $N = 59$
On the average day, about how many hours do you personally watch television?				
N	59	59	57	59
Mean (SE)	2.2 (0.24)	2.8 (0.24)	2.5 (0.28)	2.8 (0.27)
Now thinking about the time since your 18th birthday, how many male (female) partners have you had sex with?[a]				
N	16	11	16	14
Median	41.5	45.0	47.0	80.0
Do you have any credit cards or charge cards? Please do not include debit cards.				
Yes, *N*(%)	46 (78.0)	40 (66.7)	43 (75.4)	35 (59.3)
No, *N*(%)	13 (22.0)	20 (33.3)	14 (24.6)	24 (40.7)
Have you smoked at least 100 cigarettes in your entire life?				
Yes, *N*(%)	30 (50.8)	27 (45.0)	23 (40.4)	15 (25.4)
No, *N*(%)	29 (49.2)	30 (50.0)	30 (52.6)	42 (71.2)

Continued

Appendix B. *Continued*

Question items	FTF $N = 59$	HA $N = 60$	LA $N = 57$	ACASI $N = 59$
How often do you read the newspaper?				
Every day, N(%)	29 (49.2)	23 (38.3)	20 (35.1)	11 (18.6)
A few times a week, N(%)	15 (25.4)	16 (26.7)	19 (33.3)	26 (44.1)
Once a week, N(%)	4 (6.8)	9 (15.0)	8 (14.0)	6 (10.2)
Less than once a week, N(%)	10 (16.9)	7 (11.7)	5 (8.8)	13 (22.0)
Never, N(%)	1 (1.7)	5 (8.3)	5 (8.8)	3 (5.1)
Would you tell me how often you watch primetime drama or situation comedy programs?				
Every day, N(%)	7 (11.9)	11 (18.3)	6 (10.5)	11 (18.6)
Several times a week, N(%)	15 (25.4)	19 (31.7)	23 (40.4)	25 (42.4)
Several times a month, N(%)	16 (27.1)	12 (20.0)	7 (12.3)	11 (18.6)
Rarely, N(%)	14 (23.7)	13 (21.7)	14 (24.6)	6 (10.2)
Never, N(%)	7 (11.9)	5 (8.3)	7 (12.3)	6 (10.2)

Appendix B. *Continued*

Question items	FTF *N* = 59	HA *N* = 60	LA *N* = 57	ACASI *N* = 59
People have different reasons for saving, even though they may not be saving all the time. Which of the following statements comes closest to describing your saving habits?[b]				
Don't save, *N*(%)	18 (30.5)	24 (40.0)	19 (33.3)	25 (42.4)
Save occasionally, *N*(%)	24 (40.7)	19 (31.7)	26 (45.6)	26 (44.1)
Save regularly, *N*(%)	17 (28.8)	17 (28.3)	12 (21.1)	8 (13.6)
Thinking back over the last 12 months, about how regularly did you drink alcoholic beverages?				
Never in 12 months, *N*(%)	5 (8.5)	9 (15.0)	5 (8.8)	7 (11.9)
1 to 3 times in 12 months, *N*(%)	3 (5.1)	7 (11.7)	4 (7.0)	5 (8.5)
4 to 7 times in 12 months, *N*(%)	3 (5.1)	5 (8.3)	4 (7.0)	7 (11.9)
8 to 11 times in 12 months, *N*(%)	5 (8.5)	4 (6.7)	6 (10.5)	5 (8.5)
1 to 3 times a month, *N*(%)	13 (22.0)	13 (21.7)	7 (12.3)	9 (15.3)
Once or twice a week, *N*(%)	12 (20.3)	7 (11.7)	15 (26.3)	21 (35.6)
3 to 4 times a week, *N*(%)	15 (25.4)	11 (18.3)	14 (24.6)	5 (8.5)
5 times a week or more, *N*(%)	3 (5.1)	4 (6.7)	2 (3.5)	0 (0.0)

Continued

Appendix B. *Continued*

Question items	FTF $N = 59$	HA $N = 60$	LA $N = 57$	ACASI $N = 59$
During the past 12 months, how often have you offered your seat on a bus or in a public place to a stranger who was standing?				
Not at all in the past year, N(%)	7 (12.1)	4 (7.1)	3 (5.4)	7 (12.3)
Once in the past year, N(%)	2 (3.4)	9 (16.1)	5 (8.9)	6 (10.5)
At least two or three times a year, N(%)	13 (22.4)	18 (32.1)	18 (32.1)	20 (35.1)
Once a month, N(%)	15 (25.9)	8 (14.3)	11 (19.6)	12 (21.1)
Once a week, N(%)	7 (11.9)	8 (14.3)	10 (17.9)	6 (10.5)
More than once a week, N(%)	14 (23.7)	9 (16.1)	9 (16.1)	6 (10.5)
During the past 12 months, how often have you given food or money to a homeless person?				
Not at all in the past year, N(%)	14 (23.7)	13 (22.0)	9 (17.0)	16 (27.6)
Once in the past year, N(%)	4 (6.8)	10 (16.9)	6 (11.3)	4 (6.9)
At least two or three times a year, N(%)	18 (30.5)	19 (32.2)	20 (37.7)	27 (46.6)
Once a month, N(%)	13 (22.0)	8 (13.6)	9 (17.0)	10 (17.2)
Once a week, N(%)	5 (8.5)	4 (6.8)	7 (13.2)	1 (1.7)
More than once a week, N(%)	5 (8.5)	5 (8.5)	2 (3.8)	0 (0.0)

[a]Top quartile of responses, or >= 20 sex partners.

[b]Five original options have been collapsed to three options.

Supplementary Data

Supplementary data are freely available online at http://poq.oxfordjournals.org/.

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